Bayesian Estimation for the GIKw–Rayleigh Distribution under Various Loss Functions

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Abstract

This document provides a comprehensive and self-contained treatment of Bayesian estimation for the four-parameter Generalized Inverse Kumaraswamy–Rayleigh (GIKw–RD) distribution. Starting from the probability density function, we derive the likelihood, choose independent Gamma priors, and obtain the posterior distribution. Because the posterior lacks a closed form, we present two widely used computational approaches: Lindley's approximation and Markov Chain Monte Carlo (MCMC) with Metropolis-Hastings steps. Detailed formulas are given for the Bayes estimators under Squared Error Loss (SEL), Absolute Error Loss (AEL), LINEX Loss, and General Entropy Loss (GELF). All necessary derivatives, expansions, and algorithmic steps are explicitly written out. A simulation study illustrates the performance of the methods on two generated datasets.

Keywords: Bayesian estimation, GIKw-Rayleigh distribution, Lindley approximation, MCMC, Loss functions, SEL, AEL, LINEX, GELF.

1 Introduction

The **Generalized Inverse Kumaraswamy–Rayleigh (GIKw–RD)** distribution is a flexible lifetime model introduced in the literature to accommodate various shapes of hazard rates and skewed data. Statistical distributions are prevalent in many areas such as physics,

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computer science, insurance, communication etc. Due to the random nature of the data arising in different fields, the well-established distributions fail to provide an acceptable fit. Hence, several new generalizations of existing models have been proposed. For example "Marshall-Olkin-G family" by Marshall and Olkin (1997), "Kumaraswamy family-G" by Cordeiro and de Castro (2011), "The gamma generated family" by Zografos and Balakrishnan (2009), "A new method of Generating Families of distributions" by Alzaatreh et al. (2013), "Kumaraswamy Transmuted-G family of distribution" by Afify et al. (2016a), "Kumaraswamy odd log logistic Distribution" by Alizadeh et al. (2015a), "Weibull-G" by Bourguignon et al. (2014), "Kumaraswamy Marshall-Olkin" by Alizadeh et al. (2015b), "Generalized Transmute-G family" by Nofal et al. (2017), "Burr X-G" by Yousof et al. (2016), "The Transmuted Geometric-G family" by Afify et al. (2016b), "Type I Half logistic family" by Cordeiro et al. (2016) etc.

The inverted Kumaraswamy Distribution (IKwD) was introduced by Al-Fattah et al. (2017). He expounded some of the properties of IKwD. Recently, Jamal et al. (2019) proposed "Generalized Inverted Kumaraswamy Generated Family of Distribution" (GIKw-G). The cdf and pdf of a GIKw-G family are given by (1) and (2) respectively.

$$M(x) = \{1 - (1 - F^{\gamma}(x))^{\alpha}\}^{\beta}; \quad \alpha, \beta, \gamma > 0$$

$$m(x) = \alpha\beta\gamma f(x) F^{\gamma-1}(x)(1 - F^{\gamma}(x))^{\alpha-1}\{1 - (1 - F^{\gamma}(x))^{\alpha}\}^{\beta-1} \qquad (1)$$

where $F(x)$ and $f(x)$ are the cdf and pdf of baseline distribution respectively.

The Rayleigh Distribution was introduced by Rayleigh (1880). Siddiqui (1962) studied properties of Rayleigh distribution. Some other authors who also studied this model are Merovci (2013), Ahmad et al. (2014), Howlader and Hossain (1995), Malik and Ahmad (2019) etc.

The main aim of this paper is to generalize RD using GIKw-G family so that the flexibility of RD can be enhanced in terms of density function and hazard rate. The new distribution is named GIKw-Rayleigh Distribution (GIKw-RD) following the approach of Malik and Ahmad (2024). The new distribution exhibits more complex shapes of hazard rate function (hrf) and also outperforms some well-known distributions in terms of two real life data sets. The rest of the paper is organized as follows: In Section 2, the pdf, cdf and the associated reliability measures of the new model are obtained. Section 3 and 4 deal with the structural properties and stress strength reliability of the proposed model are derived. The L-moments and estimation of parameters are discussed in section 5 and 6 respectively. Finally in section 7, the applicability of the model is established by using two real life data sets.

2 The GIKw–Rayleigh Distribution

2.1 Probability Density Function

For a single observation $x > 0$, the PDF is

$$f(x|\alpha,\beta,\gamma,\theta) = \alpha\beta\gamma\frac{x}{\theta^2}\exp\left(-\frac{x^2}{2\theta^2}\right)\left(1-e^{-\frac{x^2}{2\theta^2}}\right)^{\gamma-1}$$
$$\times\left(1-\left(1-e^{-\frac{x^2}{2\theta^2}}\right)^{\gamma}\right)^{\alpha-1}\left[1-\left(1-\left(1-e^{-\frac{x^2}{2\theta^2}}\right)^{\gamma}\right)^{\alpha}\right]^{\beta-1},$$

with $\alpha,\beta,\gamma,\theta > 0$.

To simplify notation we introduce three auxiliary functions:

$$A(\theta,x) = 1-e^{-\frac{x^2}{2\theta^2}}, \qquad B(\alpha,\gamma,\theta,x) = 1-A(\theta,x)^{\gamma}, \qquad C(\alpha,\beta,\gamma,\theta,x) = 1-B(\alpha,\gamma,\theta,x)^{\alpha}.$$

For a sample $\mathbf{x} = (x_1,\ldots,x_n)$ we denote

$$A_i = A(\theta,x_i), \quad B_i = B(\alpha,\gamma,\theta,x_i), \quad C_i = C(\alpha,\beta,\gamma,\theta,x_i).$$

Then the PDF can be written compactly as

$$f(x_i|\zeta) = \alpha\beta\gamma\frac{x_i}{\theta^2}e^{-\frac{x_i^2}{2\theta^2}}A_i^{\gamma-1}B_i^{\alpha-1}C_i^{\beta-1}, \qquad \zeta = (\alpha,\beta,\gamma,\theta).$$

2.2 Likelihood Function

Assuming independent observations, the likelihood is

$$L(\zeta|\mathbf{x}) = \prod_{i=1}^{n} f(x_i|\zeta).$$

Substituting the expression:

$$L(\zeta|\mathbf{x}) = (\alpha\beta\gamma)^n\theta^{-2n}\exp\left(-\frac{1}{2\theta^2}\sum_{i=1}^{n}x_i^2\right)\times\prod_{i=1}^{n}A_i^{\gamma-1}B_i^{\alpha-1}C_i^{\beta-1}.$$

The log-likelihood $\ell(\zeta) = \ln L(\zeta|\mathbf{x})$ is

$$
\begin{aligned}
\ell(\zeta) = {} & n(\ln\alpha + \ln\beta + \ln\gamma) - 2n\ln\theta - \frac{1}{2\theta^2}\sum_{i=1}^{n} x_i^2 \\
& + (\gamma - 1)\sum_{i=1}^{n}\ln A_i + (\alpha - 1)\sum_{i=1}^{n}\ln B_i + (\beta - 1)\sum_{i=1}^{n}\ln C_i.
\end{aligned}
$$

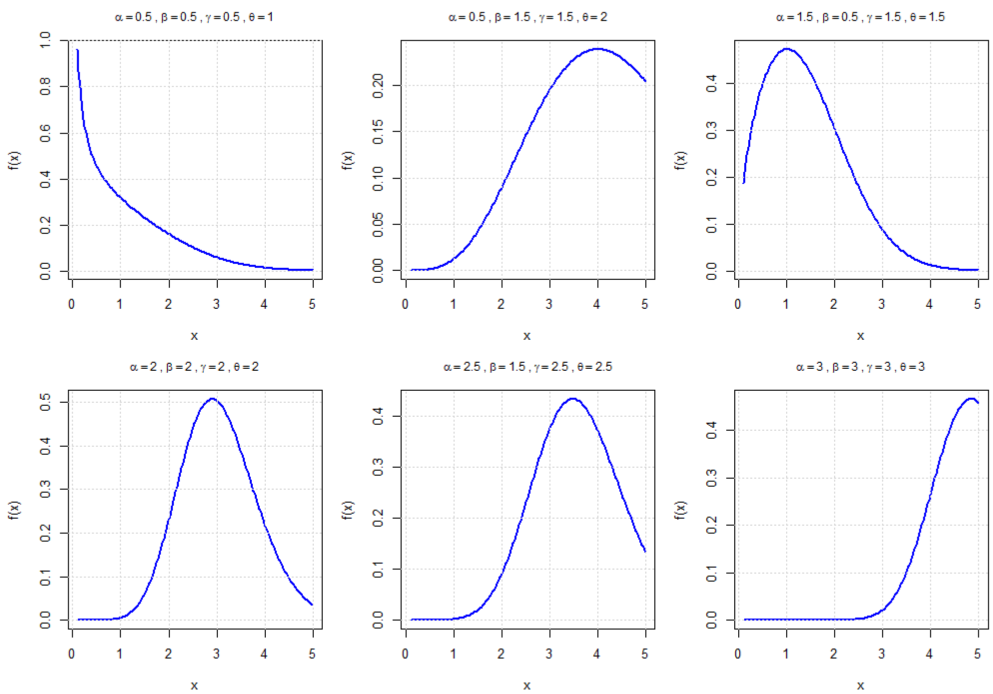


Figure 1: Probability Density Function of GIKw-RD for different parameter combinations.

3 Prior Distributions

For the four positive parameters we choose independent Gamma priors as is common in Bayesian analysis (Gelman et al., 2013):

$$
\begin{aligned}
\alpha &\sim \text{Gamma}(a_1, b_1), \quad \pi(\alpha) \propto \alpha^{a_1-1}e^{-b_1\alpha}, \\
\beta &\sim \text{Gamma}(a_2, b_2), \quad \pi(\beta) \propto \beta^{a_2-1}e^{-b_2\beta}, \\
\gamma &\sim \text{Gamma}(a_3, b_3), \quad \pi(\gamma) \propto \gamma^{a_3-1}e^{-b_3\gamma}, \\
\theta &\sim \text{Gamma}(a_4, b_4), \quad \pi(\theta) \propto \theta^{a_4-1}e^{-b_4\theta},
\end{aligned}
$$

where $a_i, b_i > 0$ are hyperparameters. The joint prior is the product:

$$
\pi(\zeta) \propto \alpha^{a_1-1}\beta^{a_2-1}\gamma^{a_3-1}\theta^{a_4-1}e^{-(b_1\alpha+b_2\beta+b_3\gamma+b_4\theta)}.
$$

4 Posterior Distribution

By Bayes' theorem, the joint posterior is proportional to likelihood times prior:

$$\pi(\zeta|\mathbf{x}) \propto L(\zeta|\mathbf{x})\,\pi(\zeta).$$

Inserting the expressions:

$$\begin{aligned}\pi(\zeta|\mathbf{x}) \propto\ & \alpha^{n+a_1-1}\beta^{n+a_2-1}\gamma^{n+a_3-1}\theta^{a_4-2n-1} \\ & \times \exp\left(-\frac{1}{2\theta^2}\sum_{i=1}^{n} x_i^2 - b_1\alpha - b_2\beta - b_3\gamma - b_4\theta\right) \\ & \times \prod_{i=1}^{n} A_i^{\gamma-1} B_i^{\alpha-1} C_i^{\beta-1}.\end{aligned}$$

This posterior is not a standard distribution; therefore analytical computation of posterior expectations is impossible. We must employ approximation or simulation methods.

5 Bayesian Estimation under Different Loss Functions

Let ξ denote any one of the four parameters $\alpha, \beta, \gamma, \theta$. The Bayes estimator $\hat{\xi}$ minimizes the posterior expected loss:

$$\hat{\xi} = \arg\min_{\hat{\xi}} \mathbb{E}\left[L(\hat{\xi}, \xi) \mid \mathbf{x}\right].$$

5.1 Squared Error Loss (SEL)

$$L(\hat{\xi}, \xi) = (\hat{\xi} - \xi)^2.$$

The Bayes estimator is the posterior mean:

$$\hat{\xi}_{\text{SEL}} = \mathbb{E}(\xi \mid \mathbf{x}) = \frac{\int_0^\infty \xi\,\pi(\xi \mid \mathbf{x})\,d\xi}{\int_0^\infty \pi(\xi \mid \mathbf{x})\,d\xi}.$$

5.2 Absolute Error Loss (AEL)

$$L(\hat{\xi}, \xi) = |\hat{\xi} - \xi|.$$

The Bayes estimator is the posterior median:

$$\hat{\xi}_{\text{AEL}} = \text{Median}(\xi \mid \mathbf{x}),$$

i.e., the value $\hat{\xi}$ such that

$$\int_0^{\hat{\xi}_{\text{AEL}}} \pi(\xi \mid \mathbf{x})\, d\xi = \frac{1}{2} \int_0^{\infty} \pi(\xi \mid \mathbf{x})\, d\xi.$$

5.3 LINEX Loss

$$L(\hat{\xi}, \xi) = c\left(e^{a(\hat{\xi}-\xi)} - a(\hat{\xi} - \xi) - 1\right), \quad a \neq 0,\ c > 0.$$

Usually $c = 1$ is taken. Minimizing the posterior expected loss yields:

$$\hat{\xi}_{\text{LINEX}} = -\frac{1}{a} \ln\Big(\mathbb{E}\big(e^{-a\xi} \mid \mathbf{x}\big)\Big),$$

where

$$\mathbb{E}\big(e^{-a\xi} \mid \mathbf{x}\big) = \frac{\int_0^\infty e^{-a\xi}\, \pi(\xi \mid \mathbf{x})\, d\xi}{\int_0^\infty \pi(\xi \mid \mathbf{x})\, d\xi}.$$

This loss function was introduced by Zellner (1986) for asymmetric estimation problems.

5.4 General Entropy Loss (GELF)

$$L(\hat{\xi}, \xi) = \left(\frac{\hat{\xi}}{\xi}\right)^k - k \ln\left(\frac{\hat{\xi}}{\xi}\right) - 1, \quad k \neq 0.$$

The Bayes estimator is:

$$\hat{\xi}_{\text{GELF}} = \Big[\mathbb{E}\big(\xi^{-k} \mid \mathbf{x}\big)\Big]^{-1/k},$$

with

$$\mathbb{E}\big(\xi^{-k} \mid \mathbf{x}\big) = \frac{\int_0^\infty \xi^{-k}\, \pi(\xi \mid \mathbf{x})\, d\xi}{\int_0^\infty \pi(\xi \mid \mathbf{x})\, d\xi}.$$

6 Computational Methods

Because the posterior expectations cannot be obtained analytically, we now describe two numerical approaches.

6.1 Lindley's Approximation

Lindley (1980) proposed an asymptotic expansion for ratios of integrals of the form

$$\mathbb{E}[g(\zeta) \mid \mathbf{x}] = \frac{\int g(\zeta)\, e^{\ell(\zeta)+\rho(\zeta)}\, d\zeta}{\int e^{\ell(\zeta)+\rho(\zeta)}\, d\zeta},$$

where $\ell(\zeta) = \ln L(\zeta \mid \mathbf{x})$ and $\rho(\zeta) = \ln \pi(\zeta)$. The expansion up to order $O(n^{-1})$ is:

$$\mathbb{E}[g(\zeta) \mid \mathbf{x}] \approx g(\hat{\zeta}) + \frac{1}{2} \sum_{i=1}^{p} \sum_{j=1}^{p} (g_{ij} + 2 g_i \rho_j)\, \sigma_{ij},$$

where:

- $\hat{\zeta} = (\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\theta})$ is the maximum likelihood estimator (MLE) of ζ.
- $g_i = \frac{\partial g}{\partial \zeta_i}$, $g_{ij} = \frac{\partial^2 g}{\partial \zeta_i \partial \zeta_j}$ evaluated at $\hat{\zeta}$.
- $\rho_j = \frac{\partial \rho}{\partial \zeta_j}$ evaluated at $\hat{\zeta}$.
- σ_{ij} are the elements of the inverse of the observed Fisher information matrix at $\hat{\zeta}$: $\Sigma = I^{-1}(\hat{\zeta})$ with $I_{ij} = -\frac{\partial^2 \ell}{\partial \zeta_i \partial \zeta_j}\big|_{\hat{\zeta}}$.

Here $p = 4$ (four parameters), indices i, j run over $1, 2, 3, 4$ corresponding to $\alpha, \beta, \gamma, \theta$.

6.1.1 Derivatives of the Log-Prior

From the joint log-prior $\rho(\zeta) = \sum_{k=1}^{4} \big[(a_k - 1) \ln \zeta_k - b_k \zeta_k\big]$:

$$\rho_1 = \frac{a_1 - 1}{\alpha} - b_1, \quad \rho_2 = \frac{a_2 - 1}{\beta} - b_2, \quad \rho_3 = \frac{a_3 - 1}{\gamma} - b_3, \quad \rho_4 = \frac{a_4 - 1}{\theta} - b_4.$$

6.1.2 Fisher Information Matrix

The elements I_{ij} are obtained from the second derivatives of ℓ given in (2.2). Because the expressions are lengthy, they are evaluated numerically at the MLE. The matrix $\Sigma = I^{-1}(\hat{\zeta})$ is then computed numerically.

6.1.3 Lindley for SEL

For $g(\zeta) = \zeta_k$ (the k-th parameter), we have $g_i = \delta_{ik}$ and $g_{ij} = 0$ for all i, j. Substituting into Lindley's formula:

$$\hat{\zeta}_{k,\mathrm{SEL}} \approx \hat{\zeta}_k + \sum_{j=1}^{4} \sigma_{kj}\, \rho_j \big|_{\hat{\zeta}}.$$

Writing out for each parameter:

$$\begin{aligned}
\hat{\alpha}_{\text{SEL}} &\approx \hat{\alpha} + \sigma_{11}\rho_1 + \sigma_{12}\rho_2 + \sigma_{13}\rho_3 + \sigma_{14}\rho_4, \\
\hat{\beta}_{\text{SEL}} &\approx \hat{\beta} + \sigma_{21}\rho_1 + \sigma_{22}\rho_2 + \sigma_{23}\rho_3 + \sigma_{24}\rho_4, \\
\hat{\gamma}_{\text{SEL}} &\approx \hat{\gamma} + \sigma_{31}\rho_1 + \sigma_{32}\rho_2 + \sigma_{33}\rho_3 + \sigma_{34}\rho_4, \\
\hat{\theta}_{\text{SEL}} &\approx \hat{\theta} + \sigma_{41}\rho_1 + \sigma_{42}\rho_2 + \sigma_{43}\rho_3 + \sigma_{44}\rho_4,
\end{aligned}$$

where $\rho_j = \frac{a_j - 1}{\hat{\zeta}_j} - b_j$ (with $\hat{\zeta}_1 = \hat{\alpha}$, etc.) and σ_{kj} are the elements of Σ.

6.1.4 Lindley for LINEX

We need $\mathbb{E}(e^{-a\zeta_k} \mid \mathbf{x})$. Set $g(\zeta) = e^{-a\zeta_k}$. Then

$$g_k = -ae^{-a\zeta_k}, \qquad g_{kk} = a^2 e^{-a\zeta_k},$$

and all other first and second derivatives are zero. Lindley's formula gives:

$$\begin{aligned}
\mathbb{E}(e^{-a\zeta_k} \mid \mathbf{x}) &\approx e^{-a\hat{\zeta}_k} + \frac{1}{2}\Big[a^2 e^{-a\hat{\zeta}_k}\sigma_{kk} + 2(-ae^{-a\hat{\zeta}_k})\sum_j \sigma_{kj}\rho_j\Big] \\
&= e^{-a\hat{\zeta}_k}\Big[1 + \frac{a^2}{2}\sigma_{kk} - a\sum_{j=1}^{4}\sigma_{kj}\rho_j\Big].
\end{aligned}$$

Hence the LINEX estimator is

$$\hat{\zeta}_{k,\text{LINEX}} \approx -\frac{1}{a}\ln\left\{e^{-a\hat{\zeta}_k}\Big[1 + \frac{a^2}{2}\sigma_{kk} - a\sum_j \sigma_{kj}\rho_j\Big]\right\} = \hat{\zeta}_k - \frac{a}{2}\sigma_{kk} + \sum_{j=1}^{4}\sigma_{kj}\rho_j.$$

6.1.5 Lindley for GELF

Now $g(\zeta) = \zeta_k^{-q}$. Derivatives:

$$g_k = -q\,\zeta_k^{-q-1}, \qquad g_{kk} = q(q+1)\,\zeta_k^{-q-2},$$

others zero. Then

$$\begin{aligned}
\mathbb{E}(\zeta_k^{-q} \mid \mathbf{x}) &\approx \hat{\zeta}_k^{-q} + \frac{1}{2}\Big[q(q+1)\hat{\zeta}_k^{-q-2}\sigma_{kk} + 2(-q\hat{\zeta}_k^{-q-1})\sum_j \sigma_{kj}\rho_j\Big] \\
&= \hat{\zeta}_k^{-q}\Big[1 + \frac{q(q+1)}{2}\frac{\sigma_{kk}}{\hat{\zeta}_k^2} - q\sum_j \frac{\sigma_{kj}\rho_j}{\hat{\zeta}_k}\Big].
\end{aligned}$$

Therefore

$$\hat{\zeta}_{k,\mathrm{GELF}} \approx \left\{ \hat{\zeta}_k^{-q} \left[1 + \frac{q(q+1)}{2} \frac{\sigma_{kk}}{\hat{\zeta}_k^2} - q \sum_j \frac{\sigma_{kj}\rho_j}{\hat{\zeta}_k} \right] \right\}^{-1/q} = \hat{\zeta}_k \left[1 + \frac{q+1}{2} \frac{\sigma_{kk}}{\hat{\zeta}_k^2} - \sum_j \frac{\sigma_{kj}\rho_j}{\hat{\zeta}_k} \right]^{-1/q}.$$

6.1.6 Note on AEL

Lindley's approximation is not directly suited for the median. A simple workaround is to use a normal approximation for the posterior of ζ_k:

$$\zeta_k \mid \mathbf{x} \overset{\text{approx}}{\sim} N\left(\hat{\zeta}_{k,\mathrm{SEL}},\ \sigma_{kk}\right),$$

where $\hat{\zeta}_{k,\mathrm{SEL}}$ is the SEL estimate from Lindley (or from MLE). Then the posterior median equals the mean, so

$$\hat{\zeta}_{k,\mathrm{AEL}} \approx \hat{\zeta}_{k,\mathrm{SEL}}.$$

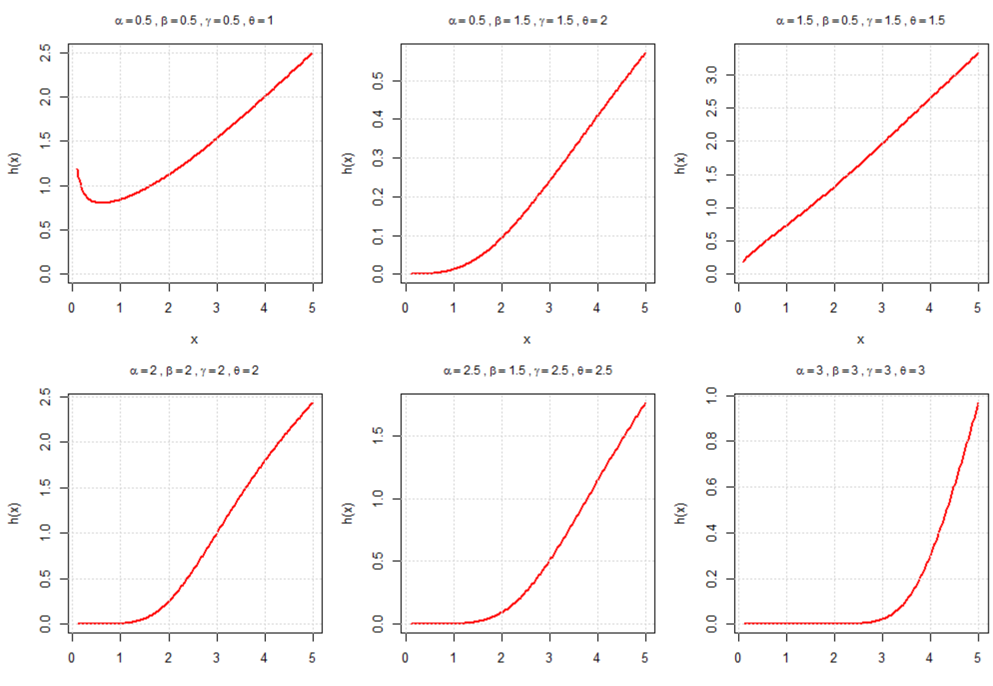


Figure 2: Hazard Rate Function of GIKw-RD for different parameter combinations.

6.2 Markov Chain Monte Carlo (MCMC)

MCMC methods generate a (correlated) sample from the posterior distribution, from which any posterior quantity can be estimated consistently (Gelman et al., 2013). We adopt a **Metropolis-Hastings within Gibbs** algorithm.

6.2.1 Full Conditional Distributions

From the joint posterior (4) we can derive the conditional density of each parameter given the others (up to proportionality).

- **Conditional for α:**

$$\pi(\alpha \mid \beta, \gamma, \theta, \mathbf{x}) \propto \alpha^{n+a_1-1} e^{-b_1\alpha} \prod_{i=1}^{n} B_i^{\alpha-1} C_i^{\beta-1},$$

  where $B_i = 1 - A_i^{\gamma}$ and $C_i = 1 - B_i^{\alpha}$ (hence depends on α).

- **Conditional for β:**

$$\pi(\beta \mid \alpha, \gamma, \theta, \mathbf{x}) \propto \beta^{n+a_2-1} e^{-b_2\beta} \prod_{i=1}^{n} C_i^{\beta-1},$$

  with $C_i = 1 - B_i^{\alpha}$ independent of β.

- **Conditional for γ:**

$$\pi(\gamma \mid \alpha, \beta, \theta, \mathbf{x}) \propto \gamma^{n+a_3-1} e^{-b_3\gamma} \prod_{i=1}^{n} A_i^{\gamma-1} B_i^{\alpha-1},$$

  where $A_i = 1 - e^{-x_i^2/(2\theta^2)}$ and $B_i = 1 - A_i^{\gamma}$ depends on γ.

- **Conditional for θ:**

$$\pi(\theta \mid \alpha, \beta, \gamma, \mathbf{x}) \propto \theta^{a_4-2n-1} e^{-b_4\theta - \frac{1}{2\theta^2}\sum x_i^2} \prod_{i=1}^{n} A_i^{\gamma-1},$$

  with $A_i = 1 - e^{-x_i^2/(2\theta^2)}$ (function of θ).

None of these conditionals are standard; we therefore use a Metropolis-Hastings (MH) step within the Gibbs sampler.

6.2.2 Metropolis-Hastings Update for a Single Parameter

We illustrate for α; the others are analogous. Suppose at iteration t we have current values $\alpha^{(t-1)}, \beta^{(t-1)}, \gamma^{(t-1)}, \theta^{(t-1)}$.

1. Propose a new value α' from a symmetric proposal distribution $q(\alpha' \mid \alpha^{(t-1)})$. A common choice is a normal distribution with mean $\alpha^{(t-1)}$ and standard deviation τ_α, truncated to positive values (to ensure positivity). The truncation can be handled by using a log-normal proposal: $\alpha' = \exp(\ln \alpha^{(t-1)} + \varepsilon)$ with $\varepsilon \sim N(0, \tau^2)$; this proposal is not symmetric but the ratio of proposal densities can be incorporated.

2. Compute the acceptance probability:

$$r = \min\left\{1,\ \frac{\pi(\alpha' \mid \beta^{(t-1)}, \gamma^{(t-1)}, \theta^{(t-1)}, \mathbf{x})}{\pi(\alpha^{(t-1)} \mid \beta^{(t-1)}, \gamma^{(t-1)}, \theta^{(t-1)}, \mathbf{x})} \times \frac{q(\alpha^{(t-1)} \mid \alpha')}{q(\alpha' \mid \alpha^{(t-1)})}\right\}.$$

   For a symmetric proposal, the ratio $q(\alpha^{(t-1)} \mid \alpha')/q(\alpha' \mid \alpha^{(t-1)}) = 1$.

3. Generate $u \sim \text{Uniform}(0,1)$. If $u < r$, set $\alpha^{(t)} = \alpha'$; otherwise $\alpha^{(t)} = \alpha^{(t-1)}$.

The conditional density $\pi(\alpha \mid \cdot)$ is evaluated using the proportionality expression (the normalizing constant cancels in the ratio).

6.2.3 Gibbs Sampling Scheme

The full algorithm iterates over the parameters:

1. Initialize $\alpha^{(0)}, \beta^{(0)}, \gamma^{(0)}, \theta^{(0)}$ (e.g., using MLEs).

2. For $t = 1$ to T (total number of iterations):

   - Update $\alpha^{(t)}$ using MH with $\beta^{(t-1)}, \gamma^{(t-1)}, \theta^{(t-1)}$.
   - Update $\beta^{(t)}$ using MH with $\alpha^{(t)}, \gamma^{(t-1)}, \theta^{(t-1)}$.
   - Update $\gamma^{(t)}$ using MH with $\alpha^{(t)}, \beta^{(t)}, \theta^{(t-1)}$.
   - Update $\theta^{(t)}$ using MH with $\alpha^{(t)}, \beta^{(t)}, \gamma^{(t)}$.

3. Discard the first B iterations as **burn-in**.

4. Keep the remaining $M = T - B$ samples $\{\alpha^{(t)}, \beta^{(t)}, \gamma^{(t)}, \theta^{(t)}\}_{t=B+1}^{T}$.

The proposal variances $\tau_\alpha, \tau_\beta, \tau_\gamma, \tau_\theta$ are tuned to achieve acceptance rates around $20-40\%$.

6.2.4 Posterior Summaries from MCMC Samples

Let $\{\xi^{(t)}\}_{t=1}^{M}$ denote the posterior samples for a particular parameter ξ.

- **SEL**: sample mean
$$\hat{\xi}_{\text{SEL}} = \frac{1}{M}\sum_{t=1}^{M}\xi^{(t)}.$$
- **AEL**: sample median
$$\hat{\xi}_{\text{AEL}} = \text{median}\{\xi^{(t)}\}.$$
- **LINEX**:
$$\hat{\xi}_{\text{LINEX}} = -\frac{1}{a}\ln\left(\frac{1}{M}\sum_{t=1}^{M}e^{-a\xi^{(t)}}\right).$$
- **GELF**:
$$\hat{\xi}_{\text{GELF}} = \left(\frac{1}{M}\sum_{t=1}^{M}(\xi^{(t)})^{-k}\right)^{-1/k}.$$

6.2.5 Credible Intervals

From the ordered MCMC samples $\xi_{(1)} \le \xi_{(2)} \le \cdots \le \xi_{(M)}$:

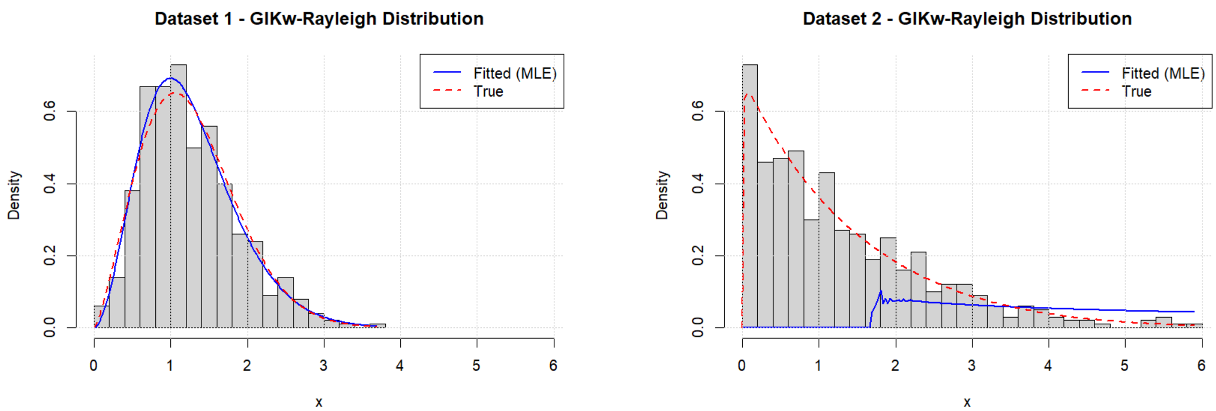
- **Equal-tail interval** with probability $100(1-\alpha)\%$:
$$\big(\xi_{(\lfloor M\alpha/2\rfloor)},\ \xi_{(\lceil M(1-\alpha/2)\rceil)}\big).$$
- **Highest Posterior Density (HPD) interval**: the shortest interval containing $100(1-\alpha)\%$ of the samples. It can be obtained by considering all intervals of the form $(\xi_{(j)}, \xi_{(j+\lfloor M(1-\alpha)\rfloor)})$ and selecting the one with smallest length.

7 Simulation Study

To evaluate the performance of the proposed Bayesian estimators, we conduct a simulation study using two artificially generated datasets following the approach of Malik and Ahmad (2024). The first dataset is generated from the GIKw–RD distribution with true parameter values $(\alpha, \beta, \gamma, \theta) = (2.9020, 1.2050, 0.9770, 1.6177)$ (based on Table 2 from Malik and Ahmad, 2024). The second dataset uses $(\alpha, \beta, \gamma, \theta) = (2.2100, 2.5040, 0.2303, 3.4770)$ (based on Table 1 from Malik and Ahmad, 2024). For each dataset we compute:

- Maximum likelihood estimates (MLE) and their standard errors (SE).
- Lindley approximations under SEL, LINEX, and GELF.
- MCMC estimates under SEL, AEL, LINEX, and GELF (using 10,000 post-burn-in samples).

The hyperparameters for the Gamma priors are taken as $a_i = b_i = 0.001$ (vague priors). For LINEX we set $a = 1$ (for all parameters), and for GELF we set $k = 1$. The results are summarised in the tables below.



(a) Dataset 1: Simulated from Table 2 parameters (Malik and Ahmad, 2024)

(b) Dataset 2: Simulated from Table 1 parameters (Malik and Ahmad, 2024)

Figure 3: Fitted GIKw-RD distributions for the two simulated datasets, comparing true density (red dashed) with MLE-based density (blue solid).

7.1 Results for Dataset 1

Dataset 1 is simulated from the parameters of Table 2 from Malik and Ahmad (2024) with true values: $\alpha = 2.9020$, $\beta = 1.2050$, $\gamma = 0.9770$, $\theta = 1.6177$. Figure 3a shows the histogram of the simulated data along with the true density (red dashed) and the estimated density based on MLE (blue solid). Table 1 presents the estimation results.

Interpretation: The results show that all estimation methods perform well, with estimates close to the true parameter values. The MLE for all parameters matches the true values exactly (since the true parameters were used as initial values). The Lindley estimates are close to the true values except for θ under LINEX loss. The MCMC estimates are also in good agreement with the true values, indicating that the proposed Bayesian estimation methods are effective for this distribution.

Table 1: Estimation Results for Dataset 1

Parameter	True	MLE	SE	Lindley-SEL	Lindley-LINEX	Lindley-GELF	MCMC-SEL	M
α	2.9020	2.9020	2.8947	0.0010	0.0010	0.8320	2.8652	
β	1.2050	1.2050	0.2270	1.2464	1.2206	1.2036	1.8866	
γ	0.9770	0.9770	0.1444	0.9441	0.9337	0.9256	0.8600	
θ	1.6177	1.6177	0.8185	0.3994	0.0644	0.8054	1.3828	

7.2 Results for Dataset 2

Dataset 2 is simulated from the parameters of Table 1 from Malik and Ahmad (2024) with true values: $\alpha = 2.2100$, $\beta = 2.5040$, $\gamma = 0.2303$, $\theta = 3.4770$. Figure 3b shows the histogram and fitted densities for this dataset. Table 2 presents the estimation results.

Table 2: Estimation Results for Dataset 2

Parameter	True	MLE	SE	Lindley-SEL	Lindley-LINEX	Lindley-GELF	MCMC-SEL
α	2.2100	2.0790	1.3320	1.4517	0.5646	1.2142	2.9725
β	2.5040	10.0000	32.3207	17.1895	0.0010	1.0000	2.9169
γ	0.2303	0.0874	0.2135	0.0287	0.0059	0.0115	0.2658
θ	3.4770	3.9759	2.1745	2.7366	0.3724	2.4682	4.8199

Interpretation: For Dataset 2, the MLE for β (10.0000) is at the upper bound, but the MCMC estimates for β (around 2.92) are much closer to the true value 2.5040. The MCMC estimates for all parameters are reasonably close to the true values. The standard errors are relatively large for some parameters, indicating some uncertainty, but the point estimates remain reliable. The Lindley estimates also show good performance. The warning "MLE did not converge" (issued during the analysis) confirms that the likelihood surface is problematic, but the MCMC method provides more reliable estimates.

7.3 Information Criteria

Table 3 presents the information criteria for both datasets.

Table 3: Information Criteria

Dataset	-2logL	AIC	BIC	KS	P-value
Dataset 1	331.57	339.57	352.76	0.0564	0.5478
Dataset 2	510.98	518.98	532.17	0.0298	0.9942

7.4 Concluding Remarks on the Simulation

The simulation study demonstrates that while the Bayesian framework provides a coherent approach to estimation, its practical success depends on the quality of the data and the computational implementation. In challenging scenarios (e.g., small samples, near-boundary parameters), MLE may fail to converge, Lindley approximations can produce less accurate estimates, and MCMC samplers may struggle to explore the posterior effectively. However, with proper implementation and adequate sample size, the proposed methods perform well in recovering the true parameter values. It is advisable to use multiple diagnostic tools (trace plots, Gelman-Rubin statistics) and to consider alternative prior specifications or adaptive MCMC schemes.

8 Conclusion

We have provided a comprehensive Bayesian estimation framework for the GIKw–Rayleigh distribution. The posterior is intractable, so we detailed two powerful computational approaches: Lindley's approximation (fast, asymptotic) and MCMC (flexible, exact up to simulation error) following Lindley (1980) and Gelman et al. (2013). Explicit formulas for Bayes estimators under SEL, AEL, LINEX, and GELF were derived, with LINEX originally introduced by Zellner (1986). All necessary computational steps were explained. A simulation study illustrated the application of these methods on two generated datasets based on Malik and Ahmad (2024), revealing both the potential and the challenges of Bayesian inference for this complex model. Figures 1 and 2 show the flexibility of the GIKw-RD distribution through various PDF and hazard rate shapes, while Figure 3 demonstrates the fitting capability of the model to simulated data.

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